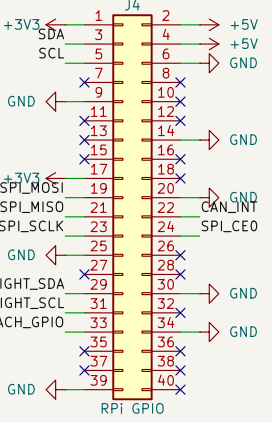
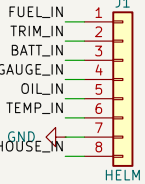


RASPBERRY PI HEADER



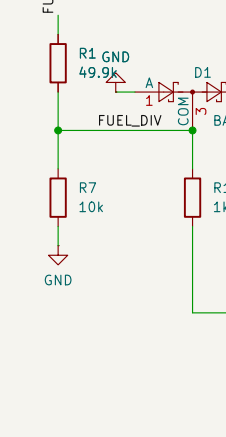
HELM: gauge taps and battery



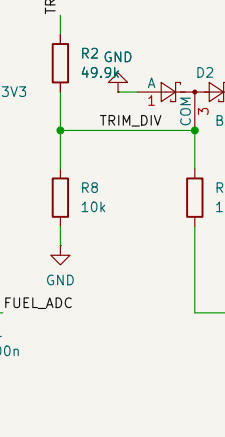
TACH: Delco EST (Ignition side)



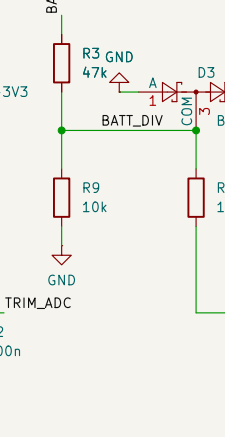
FUEL: J1.1 -> U1 AINO



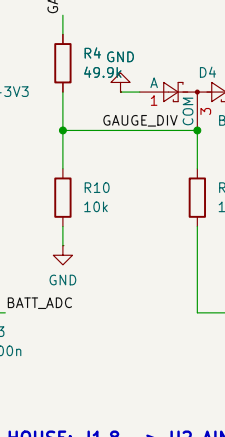
TRIM: J1.2 -> U1 AIN1



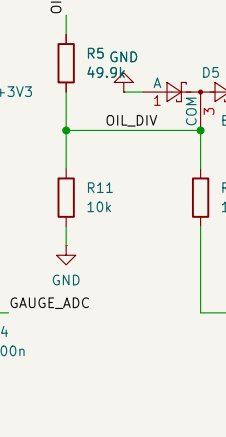
BATT: J1.3 -> U1 AIN2



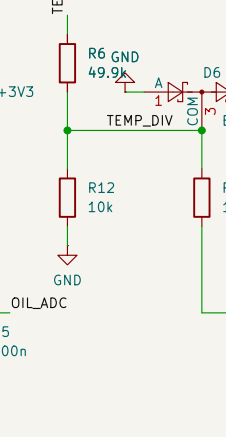
GAUGE: J1.4 -> U1 AIN3



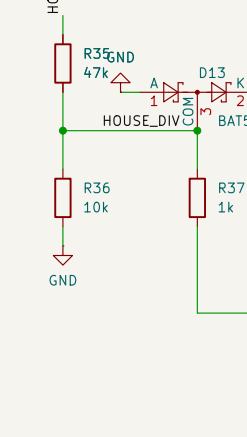
OIL: J1.5 -> U2 AINO



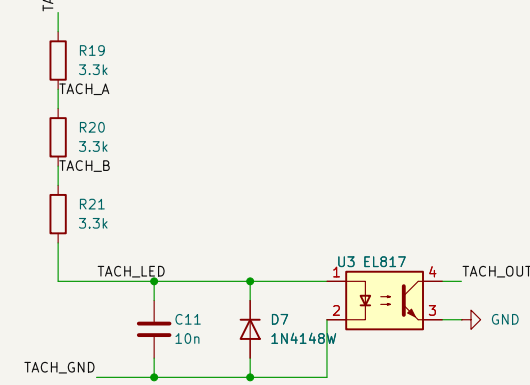
TEMP: J1.6 -> U2 AIN1



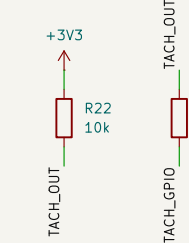
HOUSE: J1.8 -> U2 AIN2



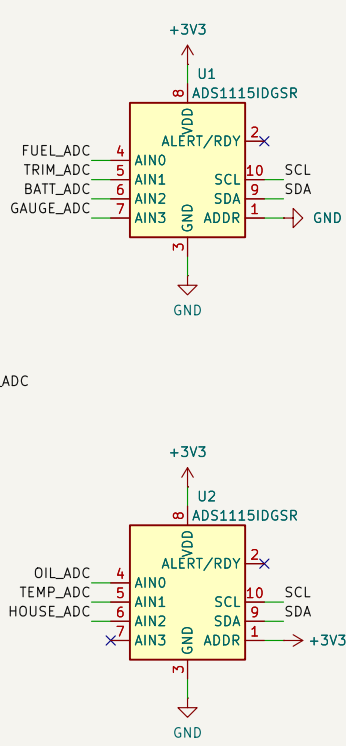
TACH: Delco EST gray wire -> optocoupler



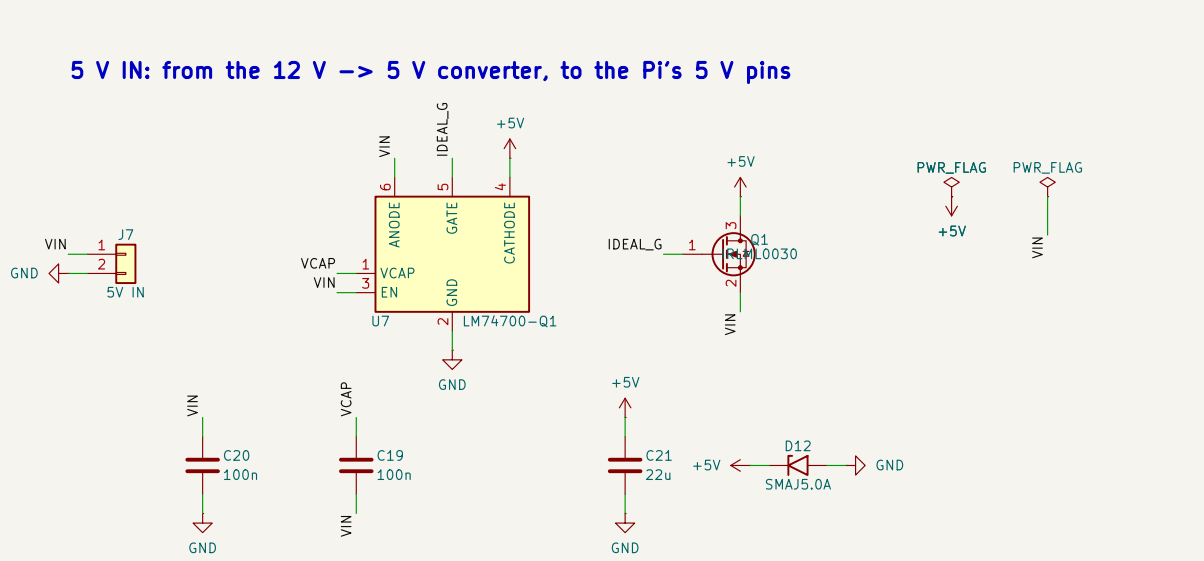
TACH OUTPUT



CONVERTERS

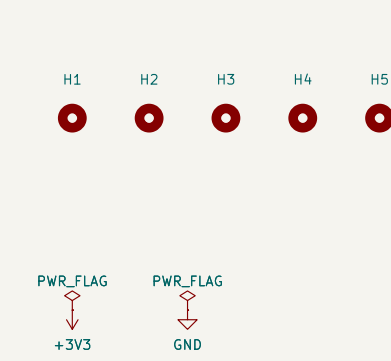


5 V IN: from the 12 V -> 5 V converter, to the Pi's 5 V pins

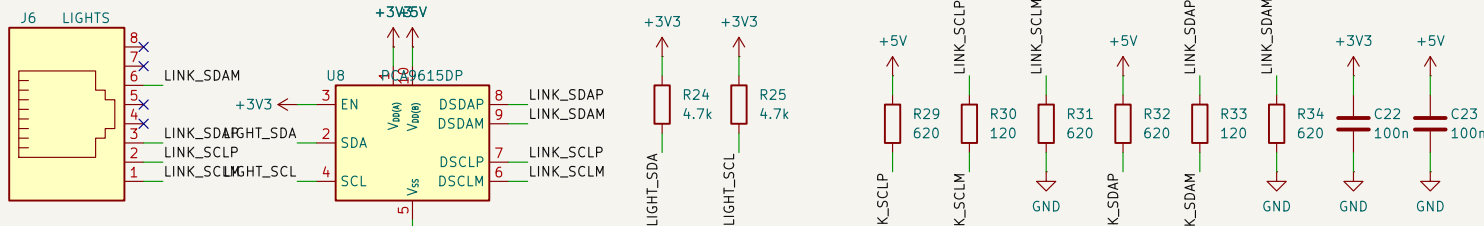


J7 takes the converter's 5 V (set 5.1–5.2 V) and runs the Pi through header pins 2 and 4; this board's 3V3 comes from the Pi. U7 + Q1 are an ideal diode (~20 mV); J7 wired backwards is blocked, and the Pi's USB-C can't feed back into the converter. D12 clamps spikes. A converter failing with 12 V out is not caught here; use one with output overvoltage protection.

MOUNTING / POWER FLAGS

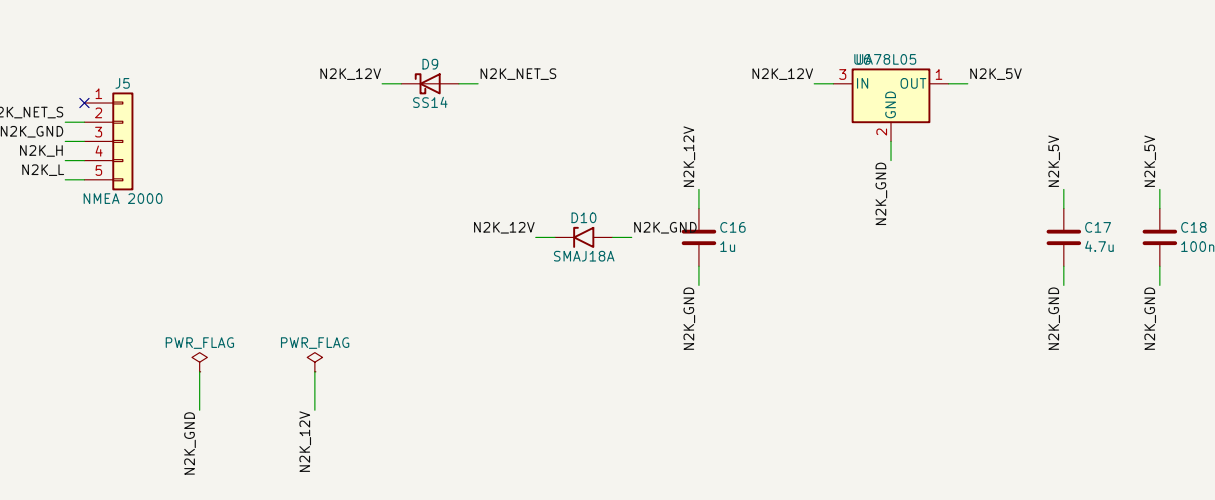


LIGHTS: the RJ45 link to the LED board (differential I2C)

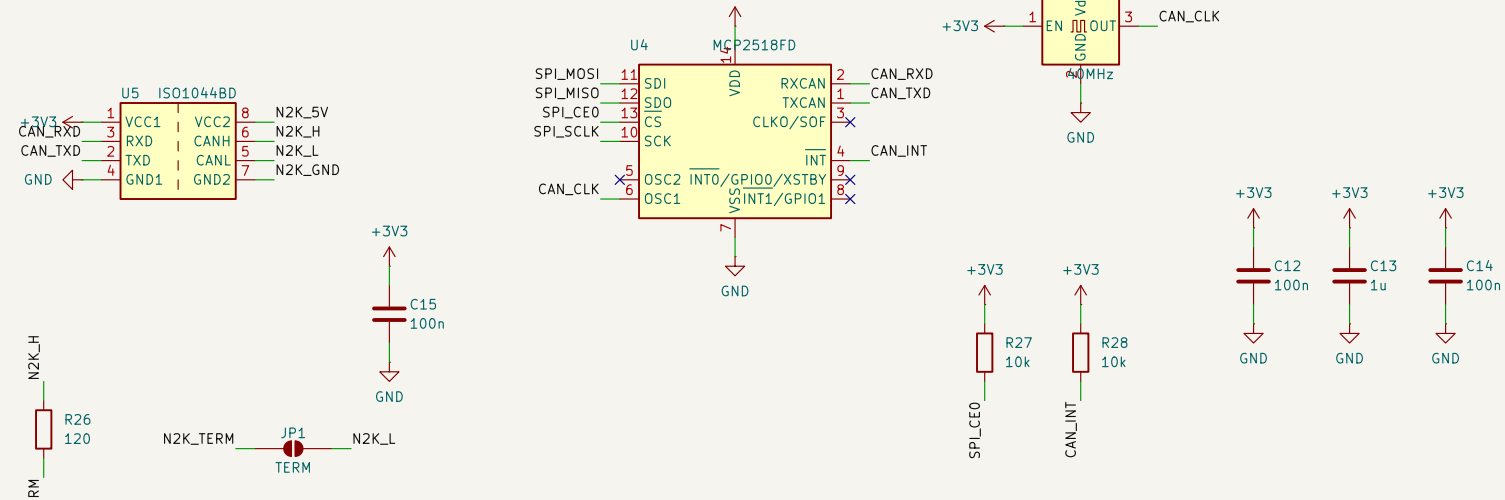


J6: a straight-through Cat5e/Cat6 patch cable to the LED board's J1 -- NOT Ethernet; never into a switch or PoE. The LED board's own I2C bus, GPIO5/6 (dtoverlay=i2c-gpio,bus=7,i2c_gpio_sda=5,i2c_gpio_scl=6), as differential I2C: pairs 1/2 SCL, 3/6 SDA, each 620/120/620 at both ends; 4/5 and 7/8 reserved. No ground or power in the cable. On it: the LED board's PCA9685 (0x40), INA226 (0x45) and RP2040 (0x30). BOAT_LED_DRIVER=ledboard, BOAT_LED_I2C_BUS=7.

NMEA 2000: network side (isolated, powered by NET-S)



NMEA 2000: PI side



NMEA 2000 (Fusion stereo, depth transducer): J5 takes a drop cable -- 1 shield (not connected), 2 NET-S, 3 NET-C, 4 NET-H, 5 NET-L. U5 isolates the network from the Pi: its network side runs from NET-S through D9 (reverse polarity), D10 (surges) and U6 (5 V), and N2K_GND is the network's 0 V, not the Pi's GND. Keep every N2K_* net 2 mm from all other copper (sensor-board.kicad.dru). About 1 LEN. JP1 (open) adds a 120 ohm terminator for a bench test with no backbone -- never bridge it on the boat; the backbone is terminated at its ends. Pi: dtparam=mcp251xfid,spi0-0,oscillator=40000000,interrupt=25 --> can0 at 250 kbit/s; BOAT_CAN=can0.

Generated from design.py by schematic.py; edit those, not this file
Passive taps on the existing analog gauges -- they stay connected and working